DEG meets SHAP: a comprehensive tool and pipeline survey

No single unified tool fully integrates DEG analysis, machine learning, and SHAP interpretation into one seamless workflow — but a rapidly maturing ecosystem of purpose-built tools, R/Python packages, and published pipelines now covers this space. The field has grown exponentially since 2022, with XGBoost + TreeSHAP emerging as the dominant combination. Benchmark studies consistently show that SHAP-derived gene sets overlap **89–99%** with traditional DEG results while producing more predictively refined biomarker panels. This survey catalogs every publicly available tool, key publication, and methodological approach connecting differential expression analysis with SHAP-based explainable ML for transcriptomics.

Purpose-built tools for ML-SHAP biomarker discovery

The most complete standalone tools that combine ML classification with SHAP-based gene ranking for omics data are described below. None internally runs DESeq2 or edgeR — all expect preprocessed expression matrices as input — but several provide end-to-end pipelines from feature selection through SHAP interpretation.

IntelliGenes (https://github.com/drzeeshanahmed/IntelliGenes) is the closest to a one-stop solution. This Python package (GitHub) trains an ensemble of seven classifiers (PubMed Central) (XGBoost, Random Forest, SVM, Logistic Regression, KNN, Gradient Boosting, Decision Trees) and computes a novel "I-Gene Score" that combines SHAP values with Herfindahl–Hirschman Index weights and ensemble feature importance. It generates SHAP swarm plots, determines biomarker directionality (over/under-expression), (Oxford Academic) and provides a cross-platform GUI. Published in *Bioinformatics* (2023, btad755) (Oxford Academic) with follow-up papers describing a GUI version (2024) and a 3D visualization extension (2025).

BiomarkerML (https://github.com/anand-imcm/proteomics-ML-workflow) offers a cloud-based automated workflow using Workflow Description Language (WDL) (bioRxiv) that runs on Terra.bio, DNANexus, or locally via Cromwell. It supports Random Forest, SVM, XGBoost, LightGBM, KNN, Logistic Regression, and deep neural networks (VAE-MLP), using **mean SHAP values** to quantify feature contributions (bioRxiv) with radar charts and summary plots. It includes protein–protein interaction network analysis for top SHAP-ranked biomarkers (bioRxiv) and weighted nested cross-validation with grid search hyperparameter tuning. Though designed for proteomics, it accepts any tabular omics data (Zhou et al., 2025, bioRxiv).

XA4C (https://github.com/QingrunZhangLab/XA4C), published in *PLOS Computational Biology* (2023), takes a distinctive approach by integrating autoencoders with TreeSHAP to evaluate each gene's contribution to latent variables learned by the autoencoder. It identifies "critical genes" that differ from traditional DEGs and hub genes, showing stronger enrichment in disease gene databases like DisGeNET and COSMIC. (PLOS) The tool supports whole-transcriptome and pathway-level analysis with DESeq2 integration and overrepresentation analysis. (GitHub)

OmicLearn (https://github.com/MannLabs/OmicLearn, http://OmicLearn.org) provides a browser-based, no-code ML tool supporting Logistic Regression, SVM, Random Forest, AdaBoost, and XGBoost for omics biomarker discovery. While it uses model-based feature importance rather than full SHAP analysis, it is widely used as the entry point for omics ML exploration (Torun et al., 2023, *J. Proteome Research*).

Tool	Language	ML Models	SHAP Type	Installation	Publication
IntelliGenes	Python	XGBoost, RF, SVM, LR, KNN, GB, DT	Core (ensemble I-Gene Score)	pip / GitHub	*Bioinformatics* 2023
BiomarkerML	Python/R/WDL	RF, SVM, XGBoost, LightGBM, KNN, LR, DNN	Mean SHAP values	WDL/Docker	bioRxiv 2025
XA4C	Python/R	Autoencoder + tree models	TreeSHAP	GitHub	*PLOS Comp Bio* 2023
OmicLearn	Python (Streamlit)	LR, SVM, RF, AdaBoost, XGBoost	Feature importance	conda/pip	*J Proteome Res* 2023
AutoGGN	Python	Graph CNNs (AutoML)	SHAP module	GitHub	*AI Life Sci* 2021

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scRNA-seq tools using SHAP for cell type markers

Several tools now apply SHAP to single-cell transcriptomics for marker gene discovery and cell type annotation, representing one of the fastest-growing applications of explainable ML in genomics.

devCellPy (https://github.com/devCellPy-Team/devCellPy) uses XGBoost classifiers with hierarchical annotation schemes for multilayered single-cell classification, achieving **>90% prediction accuracy**. Nature Its "feature ranking mode" applies the SHAP algorithm to determine positive and negative gene predictors per cell type across all annotation layers. RNA-Seq Blog Published in *Nature Communications* (2022), it validated the novel septal marker Ppp1r17 by RNAScope and demonstrated cross-species prediction from mouse to human iPSCs. Nature

scHDeepInsight (https://github.com/shangruJia/scHDeepInsight) converts gene expression matrices into 2D images via DeepInsight transformation, then uses a hierarchical CNN to classify **50 immune cell subtypes** with 93.2% average accuracy across 7 tissue datasets. bioRxiv ResearchGate SHAP-based interpretability quantifies individual gene contributions at

both base-type and subtype levels. PubMed Central PubMed Published in *Briefings in Bioinformatics* (2025). BIOENGINEER.ORG

SPmarker identifies novel cell-type marker genes in plant scRNA-seq using interpretable ML with SHAP values, demonstrating cross-species transferability (SHAP-identified markers had more orthologous genes in rice). PubMed Published in *New Phytologist* (2022). **singleDeep** (Martorell-Marugán et al., *Briefings in Bioinformatics* 2025) uses deep neural networks with SHAP for cell-type-specific feature attribution in phenotype prediction from scRNA-seq, applied to Alzheimer's and COVID-19 datasets.

NS-Forest (https://github.com/JCVenterInstitute/NSForest) deserves mention as a widely used Random Forest-based tool for minimal marker gene sets in scRNA-seq, though it uses Gini importance rather than SHAP specifically. Springer PubMed It can be extended with SHAP post-hoc.

R and Python packages enabling DEG-SHAP integration

While no single package wraps the entire DEG→ML→SHAP pipeline, a well-developed ecosystem of composable packages makes this integration straightforward.

On the R side, **ExplaineR** (`install.packages("explainer")`) published in *Bioinformatics Advances* (2024) provides SHAP-based ML interpretation for any mlr3-wrapped learner including ranger, XGBoost, and LightGBM. It offers SHAP clustering for patient subgroups, enhanced summary plots, decision curve analysis, and automated reports Oxford Academic — making it the most comprehensive R package for SHAP-based clinical/omics classification. **shapviz** (https://github.com/ModelOriented/shapviz) handles SHAP visualization with beeswarm, waterfall, force, and dependence plots, supporting XGBoost, LightGBM, H2O, and any model via kernelshap/fastshap. GitHub **kernelshap** provides model-agnostic SHAP computation in R with fast exact computation for ≤8 features. GitHub **shapr** adds conditional Shapley values that handle feature dependencies Norskregnesentral — critical for co-expressed genes. No Bioconductor package specifically integrates SHAP with DEG analysis.

On the Python side, two packages bridge the gap between R-native DEG methods and Python ML/SHAP: **PyDESeq2** (https://github.com/owkin/PyDESeq2, `pip install pydeseq2`) provides a Python implementation of DESeq2 Owkin (Muzellec et al., *Bioinformatics* 2023), while **InMoose** (https://github.com/epigenelabs/inmoose) implements limma, edgeR, and DESeq2 in Python with near-identical results to R originals. PubMed Central Combined with the `shap` library, scikit-learn, and XGBoost, these enable fully Pythonic DEG→ML→SHAP pipelines without R dependencies. GitHub

The practical recipe for a **full Python pipeline**: PyDESeq2/InMoose for DEG → filter features → train XGBoost/RF with scikit-learn → explain with `shap` TreeExplainer → visualize with `shap.summary_plot()`. For **R pipelines**: DESeq2/edgeR/limma → train via mlr3/caret → explain with kernelshap → visualize with shapviz → generate reports with ExplaineR.

Landmark publications defining the field (2020–2026)

The literature divides into benchmark studies comparing SHAP with DEG methods, disease-specific biomarker pipelines, and methodological innovations.

Yap et al. (2021, *Scientific Reports*) is the landmark validation study. Training a CNN on 18,884 genes from **16,651 GTEx samples** across 47 tissue types (Nature) (F1 = 96.1%), they used SHAP GradientExplainer to identify 2,423 discriminatory genes. (Nature) The critical finding: **98.6% of SHAP genes were also identified by edgeR DEGs**, (Nature) yet SHAP genes produced superior tissue clustering compared to all genes, DEG-selected genes, or random genes. (Nature) This established SHAP's reliability for high-dimensional RNA-seq interpretation. (Nature)

Ferrato et al. (2023, *Bioinformatics Advances*) conducted the first direct head-to-head comparison of SHAP, DEG analysis, and PCA as feature selection methods, each paired with XGBoost, LightGBM, and Random Forest on RNA-seq from 451 AML patients. (nih) **SHAP-based feature selection outperformed DEG-based selection**, with the best model (SHAP + RF) achieving 89% AUC. Their most important finding: **feature selection technique had greater impact on performance than classifier choice**.

The **MS biomarker pipeline** (arXiv 2509.22484, 2025) used Bayesian-optimized XGBoost + TreeSHAP on integrated microarray datasets, finding (arXiv) (arXiv) 167 genes overlapping between SHAP and Wilcoxon DEA, with **133 genes unique to SHAP** while DEA identified >1,000 additional genes. (arXiv) (arXiv) Both methods captured known MS genes (HLA-DRB1, HLA-DRB5), (arXiv) confirming that SHAP is more specific while DEA is more sensitive. (arXiv +2) A 2024 *BMC Genomics* study on Huntington's disease scRNA-seq demonstrated that DeepSHAP identified **low-abundance genes (MT-CO1, MT-CO2, MT-CO3) completely missed by traditional DEA** — a finding with significant implications for biomarker discovery. (PubMed +2)

Additional notable publications include cancer classification studies achieving **96.6% accuracy** across 33 TCGA cancer types with SHAP-validated biomarkers (arXiv) (arXiv) (arXiv 2410.07260), (arXiv) breast cancer deep learning + SHAP identifying DSCAM-AS1 and ESR1 as top contributors (IEEE Access, 2025), (IEEE Xplore +2) and the ABS pipeline (2026) using sequential ANOVA→Boruta→SHAP for cancer subtyping with XGBoost. (Frontiers)

Two competing architectures and when to use each

The field has converged on two distinct pipeline architectures, each with clear trade-offs.

The **pre-filtering approach** (DEG→ML→SHAP) first applies DESeq2/edgeR/limma to select statistically significant genes (typically adjusted $p < 0.05$, $|\log2FC| > 1$), then trains ML models only on this DEG subset, then applies SHAP for interpretation. This reduces dimensionality dramatically — from tens of thousands of genes to hundreds of DEGs — lowering computational cost and overfitting risk. The approach provides statistical rigor through FDR-corrected p-values and biological interpretability through fold-change magnitudes. However, because DEG

methods test each gene independently (univariate), they miss **multivariate interactions** between genes. (Frontiers) (bioRxiv) Genes with small individual effects but strong combinatorial effects get filtered out before the ML model ever sees them. Results also depend on arbitrary p-value and fold-change cutoffs. This approach is most appropriate when sample sizes are small, computational resources are limited, or the goal is mechanistically interpretable biomarkers.

The **post-hoc approach** (All genes→ML→SHAP) trains ML models on all genes without DEG pre-filtering, then uses SHAP to identify important features, optionally comparing SHAP-identified genes with DEG results afterward. This captures non-linear, multivariate interactions and eliminates arbitrary statistical thresholds. Yap et al.'s finding that 98.6% of SHAP genes overlap with DEGs while producing better clustering demonstrates that SHAP identifies a more refined subset. (Nature) The MS biomarker study showed SHAP finds **133 unique genes missed by DEA**, (arXiv) potentially representing interaction-dependent markers. (arXiv) (arXiv) The drawback is computational expense, especially with KernelSHAP on high-dimensional data, (Interpretable Machine Lear···) and increased overfitting risk with small sample sizes. (Nature) This approach works best with larger cohorts (hundreds of samples) where interaction effects are expected.

XGBoost with TreeSHAP dominates both architectures (PubMed Central) due to computational efficiency (Preprints) ($O(TLD^2)$ complexity) (Interpretable Machine Lear···) and built-in regularization for high-dimensional data. Random Forest is the second most common choice. LightGBM performs comparably but appears less frequently in published genomics studies. For deep learning models, GradientExplainer and DeepSHAP provide analogous explanations, (Springer) with a *PLOS Genetics* (2025) study confirming that **mean absolute SHAP scores and rank order of feature importance are essentially identical across TreeSHAP, KernelSHAP, and DeepSHAP** (Google APIs) — demonstrating robustness across SHAP variants.

The emerging consensus recommends **using both approaches in parallel**: DEG for statistical rigor and mechanistic insight, SHAP for predictive importance and multivariate pattern detection. Overlapping genes represent high-confidence biomarkers; SHAP-unique genes represent novel interaction-dependent candidates; DEG-unique genes are statistically robust but predictively less important. (arXiv)

Critical gaps and what does not yet exist

Several notable gaps persist in this ecosystem. **ERAPID**, confirmed as a real tool published in *Methods* (2026), is an end-to-end RNA-seq pipeline for DESeq2/dream differential expression with GSEA and literature mining (ScienceDirect) — but it contains **no SHAP or ML component**. **BioSHAP** does not exist as a tool or package in any repository searched. No **Bioconductor package** specifically integrates SHAP with DEG analysis. No **Snakemake or Nextflow pipeline** natively combines DEG with SHAP-based ML. No **dedicated web application** provides SHAP analysis for gene expression data.

Conclusion

The DEG + ML + SHAP integration landscape is best characterized as a composable ecosystem rather than a monolithic toolset. **IntelliGenes** stands as the most complete purpose-built solution for SHAP-based genomic biomarker discovery, PubMed Central while **devCellPy** and **scHDeepInsight** lead in single-cell applications. For custom pipelines, the combination of PyDESeq2 + XGBoost + shap (Python) or DESeq2 + mlr3 + kernelshap + shapviz (R) provides maximum flexibility. The key methodological insight from benchmark studies is that SHAP-based feature selection **outperforms DEG-based selection** for predictive modeling, Frontiers yet the two approaches are complementary — practitioners should run both and compare. The most important gap is the absence of a production-ready, workflow-managed (Snakemake/Nextflow) pipeline that automates the full DEG→ML→SHAP→enrichment chain, representing a clear opportunity for tool developers in this space.